

# fn sample\_geometric\_buffer

Vicki Xu (@matchaginseng), Hanwen Zhang, Zachary Ratliff (@zachratliff)

September 13, 2026

This document proves soundness of `sample_geometric_buffer` in `mod.rs` at commit `f5bb719` (outdated<sup>1</sup>).

## 1 Hoare Triple

### Precondition

#### Compiler-verified

None

#### Caller-verified

None

### Pseudocode

```
1 def sample_geometric_buffer(  
2     buffer_len: usize, constant_time: bool  
3 ) -> Optional[uint]: #  
4     if constant_time:  
5         buf = bytearray(buffer_len)  
6         fill_bytes(buf) # mutates in-place  
7         ret = None  
8         for i in range(buffer_len):  
9             # find first nonzero event  
10            if buf[i] > 0:  
11                # compute index of first nonzero bit buffer  
12                cand = 8 * i + buf[i].leading_zeroes() #  
13                ret = cand if ret is None else min(ret, cand)  
14            return ret  
15     else:  
16         for i in range(buffer_len):  
17             buf = bytearray(1)  
18             fill_bytes(buf) # mutates in-place  
19             if buf[0] > 0:  
20                 return 8 * i + buf[0].leading_zeroes()  
21  
22     return None
```

### Postcondition

**Theorem 1.1.** For any setting of the input arguments, `sample_geometric_buffer` either raises an exception if there is insufficient system entropy, or returns `sample` where `sample` is drawn from a discrete distribution.

---

<sup>1</sup>See new changes with `git diff f5bb719..0026d99 rust/src/traits/samplers/geometric/mod.rs`

`sample` is either `geo` where `geo` is a sample from the  $Geometric(p = 0.5)$  distribution, and is less than  $buffer\_len * 8$ , or `None` with probability  $2^{-buffer\_len * 8}$ .

*Proof.* `sample_geometric_buffer` uses `fill_bytes` as a subroutine to generate a buffer of `buffer_len` bytes. For each bit  $b$  in the buffer it follows that  $\Pr[b = 1] = \frac{1}{2}$  and  $\Pr[b = 0] = \frac{1}{2}$ . If there is some bit in the buffer equal to 1, the position of the *first* such bit is a zero-indexed draw from the Geometric distribution  $Geom(p)$  with  $p = 0.5$ , by definition of a Geometric random variable. If the buffer is zero, the function returns `None`.  $\square$